

Weici Pan

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About me

I am a PhD candidate in Operations Research at Stony Brook University, advised by Prof. Zhenhua Liu. My research builds on online optimization theory (competitive analysis, regret bounds, bandits) to address reliability and efficiency in AI/ML systems and, more recently, AI alignment for LLMs. I implement all methods end-to-end and evaluate them on large-scale infrastructure.

EDUCATION

PhD in Operations Research

2021 – present

Stony Brook University, Stony Brook, NY

Department of Applied Mathematics and Statistics

Advised by Prof. Zhenhua Liu

Research direction: Reliable and Efficient Systems for AI/Machine Learning

Preliminary exam committee: Prof. Zhenhua Liu, Prof. Jian Li, Prof. Adam Wierman (Caltech), Dr. Yuan Chen (NVIDIA)

BEng in Computer Science and Technology (Yao Class)

2017 – 2021

Tsinghua University, China

Yao Class, Institute for Interdisciplinary Information Sciences

Advised by Prof. Zhixuan Fang

EXPERIENCE

Visiting Student Researcher

2020

California Institute of Technology, Department of Computing and Mathematical Sciences

Mentored by Prof. Adam Wierman

Project topic: online control with additive and multiplicative noises

Junior Research Assistant

2019

The Chinese University of Hong Kong, Department of Information Engineering

Mentored by Prof. Minghua Chen

Project topic: energy-related scheduling via optimization with an inventory

PUBLICATIONS

Conference

Nie C, Pan W, Fonseca R, Liu Z. Aladdin: Joint Placement and Scaling for SLO-Aware LLM Serving. *MAS-COTS 2026*

Pan W, Barron X, Zhou J, Liu Z. Spillover-Aware Multi-Value Steering for Pluralistic LLM Alignment. *COLM 2026*

Pan W, Liu Z. Rollback Is Not Undo: Path-Dependent Failures in LLM-Arbitrated Network Control. *The 8th International Workshop on Intelligent Cloud Computing and Networking (ICCN 2026)*

Fu H, Pan W, Liu Z, Lin S. MALLM: Multi-Agent Decision-Making with LLMs for Multi-User Edge-Sensor Environments. *IEEE AIoT 2025*

Pan W, Liu Z. Switching Constrained OCO with Predictions and Feedback Delays. *IFIP Performance 2025*

Pan W, Shi G, Lin Y, Wierman A. Online Optimization with Feedback Delay and Nonlinear Switching Cost. *ACM SIGMETRICS 2022*

Journal

Pan W, Liu Z. Switching constrained OCO with predictions and feedback delays. *Performance Evaluation*. 2025 Nov 5:102524. [Journal Version of the paper at IFIP Performance 2025]

Fu H, Pan W, Zhou L, Zhang Z, Liu Z, Lin S. MALLM: Multi-Agent Decision-Making with LLMs for Multi-User Edge-Sensor Environments. *ACM SIGMETRICS Performance Evaluation Review*. 2025 Aug 27;53(2):3-8. [Journal Version of the paper at AI Crossroads Workshop]

Pan W, Liu Z. Non-stationary Bandits with Heavy Tail. *ACM SIGMETRICS Performance Evaluation Review*. 2024 Sep 6;52(2):33-5. [Journal Version of the paper at MAMA 2024]

Pan W, Liu Z. Switching Constrained Online Convex Optimization with Predictions and Feedback Delays. *ACM SIGMETRICS Performance Evaluation Review*. 2023 Oct 2;51(2):3-5. [Journal Version of the paper at MAMA 2023]

Pan W, Shi G, Lin Y, Wierman A. Online optimization with feedback delay and nonlinear switching cost. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*. 2022 Feb 25;6(1):1-34 [Journal Version of the paper at ACM SIGMETRICS 2022].

Others

Han BS, Pan W, Tay YC. Industrial Panel Discussion: The Impact of AI/ML on SIGMETRICS. *ACM SIGMETRICS Performance Evaluation Review*. 2025 Aug 27;53(2):143-4.

Pan W, Nie C, Liu, Z. Efficient Federated Learning on Edge Devices over Heterogeneous Network and Data. Poster at *OSDI 2023*

Invited Talks

Failure-Aware Dynamic Reserve and Checkpoint Scheduling for Sustainable GPU Clusters

2025 INFORMS Annual Meeting, Atlanta, GA, October 2025.

Presented a probabilistic framework co-optimizing operational cost and CO₂ emissions for GPU clusters under stochastic failures.

Works Under Review / In Progress

Aladdin: Joint Placement and Scaling for SLO-Aware LLM Serving

Joint GPU placement and autoscaling under per-request SLOs, achieving up to 71% cost reduction via stage-level latency modeling.

From Packets to Prompts: Flow Control for the LLM-Centric Web

Feedback-driven flow control generalizing congestion control to heterogeneous LLM resource vectors (GPU utilization, queue delay, memory), with stability guarantees.

Cannikin: Training DNN Models over Heterogeneous Clusters

PyTorch-based data-parallel training system with adaptive batch-size scheduling for heterogeneous GPU clusters.

Teaching

AMS 691.01: Recent Progress in AI/ML: Applications, Architectures, and Systems

Teaching Assistant Fall 2025

AMS, Stony Brook University

Co-teaching the course, giving lectures on Transformers, AI Hallucinations, and AutoML.

AMS 303 GRAPH THEORY

Teaching Assistant Fall 2021 and Spring 2022

AMS, Stony Brook University

Services

COLM 2026

Volunteer

IEEE INFOCOM 2026

Reviewer

Delegated as the reviewer of papers submitted to IEEE INFOCOM 2026.

ACM SIGMETRICS 2025

Web Chair

Responsible for setting up and maintaining the conference webpage, in collaboration with Sri Pramodh Rachuri

SKILLS

- **Programming:** Python, C++, Mathematica
- **ML/AI:** PyTorch, Transformers (Hugging Face), OpenAI/DeepSeek API, scikit-learn
- **Systems & Simulation:** NS-3, Docker, Slurm
- **Data & Optimization:** NumPy, pandas, Gurobi, CVX
- **Tools:** Git, Jupyter, $\text{\LaTeX}$